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La vita è una tempesta,
e prederla nel culo è un lampo

Life is a storm,
and getting fucked strikes like lightning.

A. Bazzani

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Introduction

Your introduction text here.

Chapter 1

Theoretical Background

1.1 Optical Dipole Trap

Optical traps are experimental tools that allow the confinement of atom or ions within a localized region of space, under the condition that these particles have a kinetic energy lower than the trap potential depth (which is usually measured in Kelvin). For the work presented in this thesis, among all possible trapping techniques, the focus will be strictly placed on Optical Dipole Traps (ODTs) for neutral atoms

Optical dipole traps rely on the electric dipole interaction of the atoms with far-detuned light. Following the work of Grimm et al. [1], this section reviews the theory behind the working mechanism of this technique, particularly in the case of red-detuned ODTs.

1.1.1 Oscillator model

The optical dipole force arises from the interaction of the induced atomic momentum with the intensity gradient of the light field. It is a conservative force, thus in the following its potential energy will be derived modelling the atom as an oscillator subjected to a classical oscillating electric field.

1.1.1.1 Interaction between induced dipole moment and driving field

The laser field is modeled as the oscillating electric field:

$$\mathbf{E}(\mathbf{r}, t) = \hat{\mathbf{e}}\mathcal{E}(r)e^{-i\omega t} + c.c. \ , \tag{1.1}$$

where $\hat{\mathbf{e}}$ is the unitary polarization vector and $\mathcal{E}(r)$ is the complex amplitude of the field. Denoting the atomic polarizability as α, the induced dipole momentum of the atom is given by:

$$\mathbf{p}(\mathbf{r}, t) = \alpha\mathbf{E}(\mathbf{r}, t) = \hat{\mathbf{e}}\alpha\mathcal{E}e^{-i\omega t} + c.c. \tag{1.2}$$

In the semiclassical framework, the interaction of a neutral atom with a laser light field can be described using the dipole approximation. The interaction Hamiltonian takes the form $H_{\text{dip}} = -\mathbf{p} \cdot \mathbf{E}$. For an induced dipole, where the dipole moment depends linearly on the external field, the potential energy is obtained by integrating the interaction over the charging of the dipole:

$$U_{\text{dip}}(t) = \int_0^{\mathbf{E}} -\alpha \mathbf{E}' \cdot d\mathbf{E}' = -\frac{1}{2}\alpha E^2 = -\frac{1}{2}\mathbf{p} \cdot \mathbf{E}\,. \tag{1.3}$$

Averaging over a field period [1]:

$$U_{\text{dip}} = -2\langle \mathbf{p} \cdot \mathbf{E} \rangle = -\frac{1}{2\epsilon_0 c}\text{Re}[\alpha] I, \tag{1.4}$$

where $I = 1/2\epsilon_0 c|\mathcal{E}|^2$ is the time-averaged field intensity.
This result shows that an atom in an oscillating laser field experiences a force proportional to the field intensity I, and to the real part of the polarizability which represent the in-phase component of the dipole oscillation, responsible for the dispersive properties of the interaction.

It follows that the dipole force is a conservative force proportional to the electric field intensity gradient:

$$\mathbf{F}_{\text{dip}}(\mathbf{r}) = -\nabla U_{\text{dip}} = \frac{1}{2\epsilon_0 c}\text{Re}[\alpha]\nabla I(\mathbf{r}). \tag{1.5}$$

The dipole oscillation absorbs a power

$$P_{\text{abs}} = \langle \dot{\mathbf{p}}\mathbf{E} \rangle = 2\omega \text{Im}[\alpha \mathcal{E}^2] = \frac{\omega}{2\epsilon_0 c}\text{Im}[\alpha] I, \tag{1.6}$$

which is proportional to the out-of-phase component of the oscillating dipole. The absorbed power can be seen as series of absorption and spontaneous emission of photons with energy $\hbar\omega$. Therefore, the total scattering rate is given by

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{P_{\text{abs}}}{\hbar\omega} = \frac{1}{\hbar\epsilon_0 c}\text{Im}[\alpha] I(\mathbf{r}). \tag{1.7}$$

1.1.1.2 Atomic polarizability

The polarizability α can be calculated by considering the classical Lorentz oscillator model for the atom, according to which the polarizability can be calculated by solving the differential equation

$$\ddot{x} + \Gamma(\omega)\dot{x} + \omega_0^2 x = -\frac{eE(t)}{m_e}, \tag{1.8}$$

Using $E(t) = \mathcal{E}_0 \exp(-i\omega t)$, thus looking for a solution in the form $x(t) = x_0 \exp(-i\omega t)$, equation (1.8) gives

$$x_0 = -\frac{e\mathcal{E}_0}{m_e} \frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}. \tag{1.9}$$

Using $p = -ex(t) = -ex_0 \exp(-i\omega t)$ and (1.9), one can easily obtain the polarizability

$$\alpha(\omega) = \frac{e^2}{m_e} \frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}, \tag{1.10}$$

where $\Gamma(\omega)$ is the classical damping rate due to the radiative energy loss.

$$\Gamma^{\text{class.}}(\omega) = \frac{e^2\omega^2}{6\pi\epsilon_0 m_e c^3}. \tag{1.11}$$

Defining the on-resonance spontaneous emission rate $\Gamma \equiv \Gamma^{\text{class.}}(\omega_0) = (\omega_0/\omega)^2\Gamma^{\text{class.}}(\omega)$, the polarizability becomes

$$\alpha(\omega) = 6\pi\epsilon_0 c^3 \frac{\Gamma/\omega_0^2}{\omega_0^2 - \omega^2 - i(\omega^3/\omega_0^2)\Gamma}. \tag{1.12}$$

In a quantum two level system subjected to classical radiation, Γ should be substituted with the spontaneous emission rate

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3} |\langle e| \hat{d} |g\rangle|^2, \tag{1.13}$$

where $\hat{d}$ is the dipole moment operator.

Optical dipole traps works in the regime of far-detuden low intensity light, thus there is no need to calculate the quantum result, and use the classical result instead [1].

1.1.1.3 Dipole potential and scattering rate

Substituting α form (1.12) in (1.4) and (1.7) one gets the expressions of the dipole potential and the scattering rate of a dipole trap as explicit function of ω, ω_0 and Γ:

$$U_{\text{dip}}(\mathbf{r}) = -\frac{3\pi c^2}{2\omega_0^3} \left(\frac{\Gamma}{\omega_0 - \omega} + \frac{\Gamma}{\omega_0 + \omega} \right) I(\mathbf{r}), \tag{1.14}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3} \left(\frac{\omega}{\omega_0} \right)^3 \left(\frac{\Gamma}{\omega_0 - \omega} + \frac{\Gamma}{\omega_0 + \omega} \right)^2 I(\mathbf{r}). \tag{1.15}$$

These equations show two resonance frequency for $\omega = \pm\omega_0$. Introducing the detuning $\Delta = \omega - \omega_0$, for the far-red regime it holds that $\Gamma \ll |\Delta|$. For instance, the D_2 line in Rb atoms has a transition frequency of ≈ 3.8 THz thus a trap works generally at few Thz of detuning, while $\Gamma \approx 36$ MHz. This allows the introduction of the well-known rotatingwave approximation, according to which the anti-resonant term can be neglected, further simplifying the equations to

$$U_{\mathrm{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I(\mathbf{r}) \tag{1.16}$$

$$\Gamma_{\mathrm{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\Gamma}{\Delta}\right)^2 I(\mathbf{r}) = \frac{\Gamma}{\hbar\Delta}U_{\mathrm{dip}} \tag{1.17}$$

Equations (1.16) and (1.17) are the fundamental equations governing the behavior of optical dipole traps and yield important physical considerations for their implementation. The sign of U_{dip} is determined by the sign of the detuning Δ. For red detuning ($\Delta < 0$) the potential is negative, confining atoms at intensity maxima, whereas for blue detuning ($\Delta > 0$), the potential is positive pushing atoms to intensity minima. Because the potential depth scales as I/δ while the scattering rate scales as I/Δ^2, optical traps are typically realized with high intensity field and large detunings, in this way, the trap depth is maximized while maintaining a low scattering rate.

1.1.2 Multi-level atoms

Moving from the two-level atom to a real atom, the electronic transitions become more complex, and the calculation of the dipole potential must take into consideration all the possible allowed transition. In particular this lead to a dipole potential that depends on the particular sub-state of the atom.

1.1.2.1 Ground-state light shifts

Labelling as ε_i the energy levels of the multi-level atom, the effect of the laser light on the levels can be described by means of the non-degenerate time-independent second order perturbation theory. Considering the interaction Hamiltonian $\mathcal{H}_I = -\hat{d}\mathbf{E}$, the perturbative correction on the i-th energy level is

$$\Delta E_i = \sum_{j\neq i}\frac{|\langle j|\mathcal{H}_I|i\rangle|}{\varepsilon_i - \varepsilon_j}. \tag{1.18}$$

The calculation can be developed by employing the dressed atom approach, in which the considered quantum states are a combination of one atomic state plus the laser field state. Defining as zero the energy of the unperturbed atomic state, in this framework an unperturbed dressed state has energy $\varepsilon_i = n\hbar\omega$ that depends only on the number n of photons in the field. The absorption of a photon by the atom causes the transition to another dressed state characterized by the energy $\varepsilon_j = \hbar\omega_j + \hbar(n-1)\hbar\omega$, where $\hbar\omega_j$ is the energy of the atomic excited state. In such a way the denominator of equation (1.18)

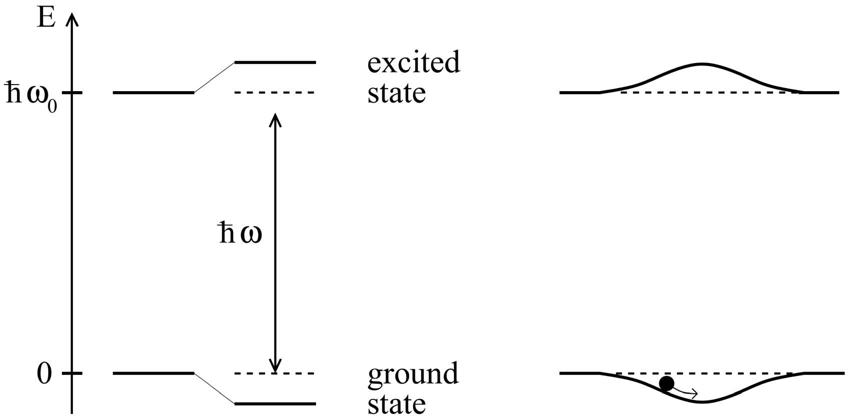

Figure 1.1: From [1]. Left: light-shifted energy levels for a two-level atom for a red detuning ($\Delta < 0$). Right: a spatially inhomogeneous field produces a potential well that traps the atoms.

becomes

$$\varepsilon_i - \varepsilon_j = -\hbar\omega_j + \hbar\omega = -\hbar\Delta_{ij}, \tag{1.19}$$

where Δ_{ij} is the detuning of the transition with respect to the laser field.

Considering a two-level atom first, (1.18) has only one term, using equations (1.13) and (1.19) and $I = 2\epsilon_0 c|\mathcal{E}|^2$, one gets:

$$\Delta E = \pm\frac{|\langle e|\mu|g\rangle|^2}{\Delta}|E|^2 = \pm\frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I, \tag{1.20}$$

for the ground and excited state (upper and lower sign, respectively). Equation (1.20) shows that the energy shift of the ground state (the solution with the sign +) exactly correspond to the dipole potential (1.16). This allows the reinterpretation of the semi-classical result obtain for the external degrees of freedom (position and momentum) in terms of the dynamic of the internal degrees of freedom. In presence of the laser field, the energy levels are shifted by a quantity proportional to the field intensity with a sign that depends on the sign of the detuning. If a low saturation regime is assumed ($I/\Delta \ll 1$), all the atom can be considered in the ground state, and the light-shifted ground state became the effective potential that govern the motion of the atoms which drives them toward region of minima of ΔE (Figure 1.1).

For a multi-level atom equation (1.18) contains many terms similar to equation (1.20), accounting for all the allowed transitions from any ground state sublevel $|g_i\rangle$ to any excited state $|e_j\rangle$. However, each of these terms depends on the dipole matrix element $d_{ij}\ \langle g_i|\hat{d}|e_j\rangle$. Such quantity can be expressed as

$$d_{ij} = c_{ij}||d|| \tag{1.21}$$

where $||d||$ is the reduced matrix element given by (1.13) and c_{ij} are real transition coefficients that take into account the coupling strength of the two involved states $|g_i\rangle$ and $|e_j\rangle$, and depends on their angular momentum value [1][2].
Therefore, the energy shift (1.18) for the ground state $|g_i\rangle$ is

$$\Delta E_i = \frac{3\pi c^2 \Gamma}{2\omega_0^3} I \times \sum_j \frac{c_{ij}^2}{\Delta_{ij}} \tag{1.22}$$

and according to the interpretation given from the two-level atoms, the dipole potential at low saturation is

$$U_{\text{dip}} = \Delta E_i. \tag{1.23}$$

Thus, the dipole potential in a multi-level atom arises from the contribution of all the allowed transition, each weighted by the ratio between its line strengths c_{ij}^2 and its detuning Δ_{ij}.

1.1.2.2 Fine and Hyperfine structures contribution for Alkali atoms

Many laser cooling experiments utilize alkali metal atoms. This preference arises because their principal transition frequencies fall within the VIS-IR range making them easily accessible with standard laser technologies. Moreover, their relatively simple atomic structure are particularly convenient, allowing easy identification of closed transitions necessary for continuous cooling cycles while minimizing the number of repumper need to address the population leaking to dark states.
Considering the fine structure arising from the electric spin-orbit coupling, the alkali atoms are found in the electronic ground state ${}^2S_{1/2}$, from this state the dipole transition rules ($\Delta L = \pm 1$, $\Delta S = 0$) allow two transition called $D-$lines, in particular:

- D_1: ${}^2S_{1/2} \rightarrow {}^2P_{1/2}$;
- D_2: ${}^2S_{1/2} \rightarrow {}^2P_{3/2}$.

Then the energy levels are further split by the coupling between the nuclear spin and the electron total momentum, the resulting structure is called hyperfine structure. Labelling $\hbar\Delta_{FS}$ the splitting due to the fine structure, $\hbar\Delta_{HFS}$ the energy splitting of the ground state hyperfine sublevel and $\hbar\Delta'_{HFS}$ the excited state hyperfine split, it holds that $\Delta_{FS} \gg \Delta_{HFS} \gg \Delta'_{HFS}$. The electronic energy level scheme for an alkali atom with $I = 3/2$ is shown in figure 1.2. Assuming that all optical detunings stay large compared with the excited-state hyperfine splitting Δ'_{HFS}, equation (1.22) can be calculated for an alkali atom in the ground state of total angular momentum F and magnetic quantum number m_F, obtaining [1]:

$$SwagBarca \tag{1.24}$$

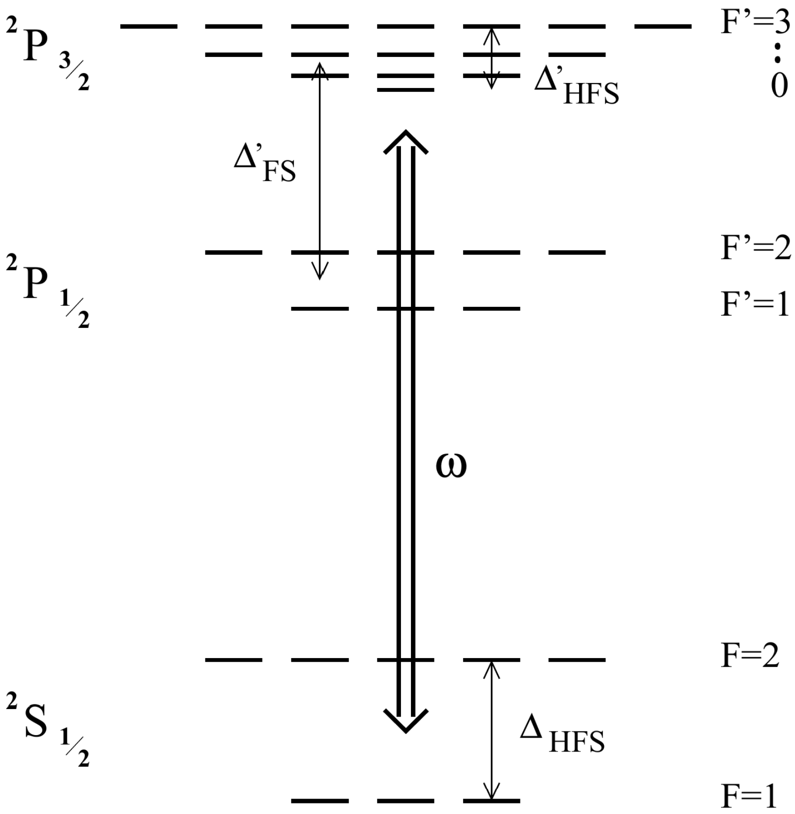

Figure 1.2: From [1]. Energy level scheme of an alkali atom with $I = 3/2$ such as ^{87}Rb.

where the two term in the square bracket represent respectively the contribution of the D_2 and D_1 line. g_F is the Landé factor, $\mathcal{P}$ characterizes the laser polarization ($\mathcal{P} = 0, \pm 1$ for linear polarization and circular $\sigma^{\pm}$ respectively), and $\Delta_{1,F}$ and $\Delta_{2,F}$ represent the energy splitting between the ground state $^2S_{1/2}, F$ and the center of the hyperfine splitting of respectively the excited states $^2P_{1/2}$ and $^2P_{3/2}$.

1.2 Optical Fibers

Bibliography

[1] Rudolf Grimm, Matthias Weidemüller, and Yurii B. Ovchinnikov. *Optical dipole traps for neutral atoms*. 1999. arXiv: physics/9902072 [physics.atom-ph]. URL: https://arxiv.org/abs/physics/9902072.

[2] Harold J. Metcalf and Peter Straten. *Laser Cooling and Trapping*. Graduate Texts in Contemporary Physics. Springer New York, NY, 1999. DOI: https://doi.org/10.1007/978-1-4612-1470-0.